Expanding the Utility of Supercritical Water Oxidation for PFAS Solid Waste Treatment

Alexis Wells Carpenter, PhD[1], Edward Hart[1], Nadezda Ojeda[1], Eric Wu[1], Ken Liberty[2], John Follin[2]
[1]AxNano; [2]General Atomics

Air Waste Management Association
PFAS Remediation Conference
Minneapolis, Minnesota
May 2025

© 2025 AxNano Inc

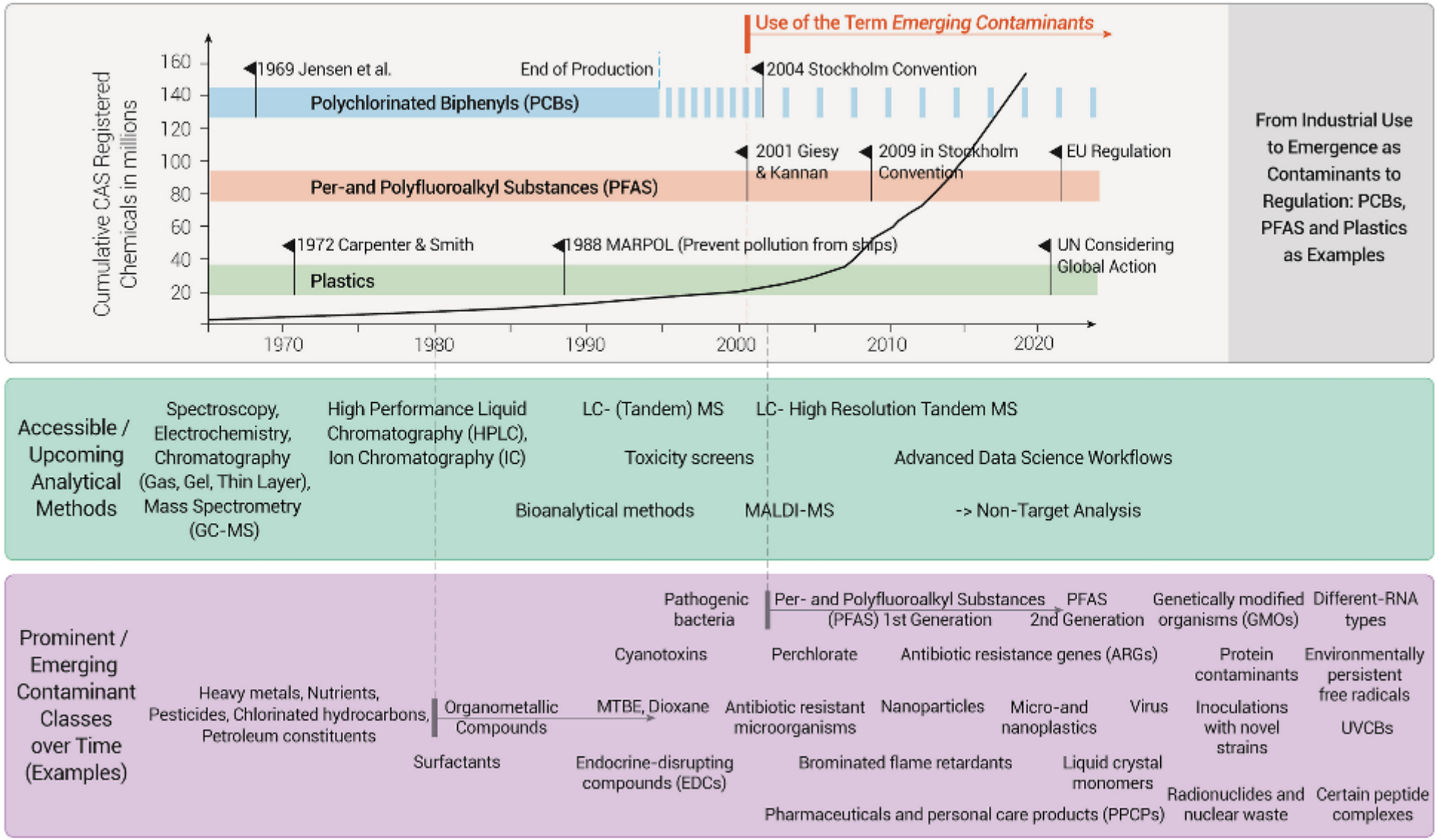
HARDWARE SOLUTIONS FOR PFAS CAPTURE AND DESTRUCTION

- Advanced, military-vetted and deployed
- Post-concentration destruction: spent sorption media, concentrates, brines
- Hardware as a Service (HaaS) model lowers cost, complexity of acquisition
- Customer traction with Army, Air Force, private industry
- Partnership with General Atomics

GENERAL ATOMICS

AxNano

© 2025 AxNano Inc



Wang et al. *The Innovation*, 2024, 5(4), 100612.

© 2025 AxNano Inc

FEASIBILITY TESTING ON SCWO COMPLETED ON A WIDE RANGE OF WASTE TYPES

Waste Stream	Contaminant of Concern (COC)	COC Conc Before Treatment	% Reduction
Motor Oil	1,4-Dioxane	1000 ppm	**>99.999%**
Surface Water	Microplastics	7%	**>99.999%**
Firefighting Foams	PFAS	~974 ppm	**~99.999%**

Demonstrated Destruction of Waste Materials by General Atomics' iSCWO:

- Pesticides
- Waste paint, oils, and lubricants
- Reverse osmosis reject
- Polychlorinated biphenyls (PCB)
- Organic solvents
- Plastic waste
- Energetic materials
- Fire retardants

GENERAL ATOMICS AxNano

© 2025 AxNano Inc

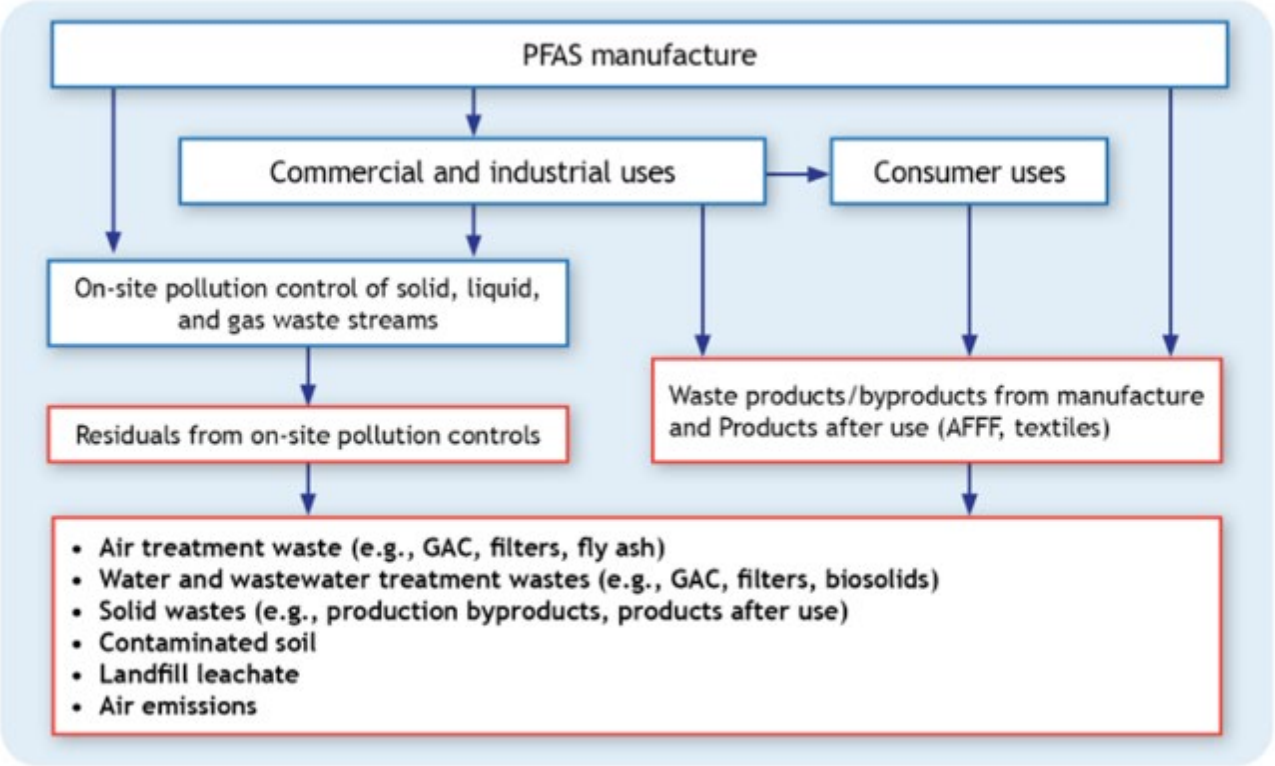
BOUNDARY CONDITIONS FOR WASTE FEED AND REACTOR CONDITIONS ENSURE OPTIMAL PERFORMANCE

- Particle size
- Solids to liquids ratio (pumpability)
- Salt concentration and speciation
- Miscibility
- Calorific value (BTU/lb)
- Feed Rate
- Compatibility
- Reactor conditions
- Contaminant of concern concentration

GENERAL ATOMICS

AxNano

© 2025 AxNano Inc



Note: The red-outlined portions show where the FY 2020 NDAA material types occur in the course of manufacture, use, and disposal of PFAS and PFAS-containing materials that are within the scope of this interim guidance.

Source: EPA Interim Guidance on PFAS Disposal 2024.

PFAS SOLIDS MANAGEMENT IS NEEDED IF INCINERATION AND LANDFILL ARE TO BE BANNED

AxNano

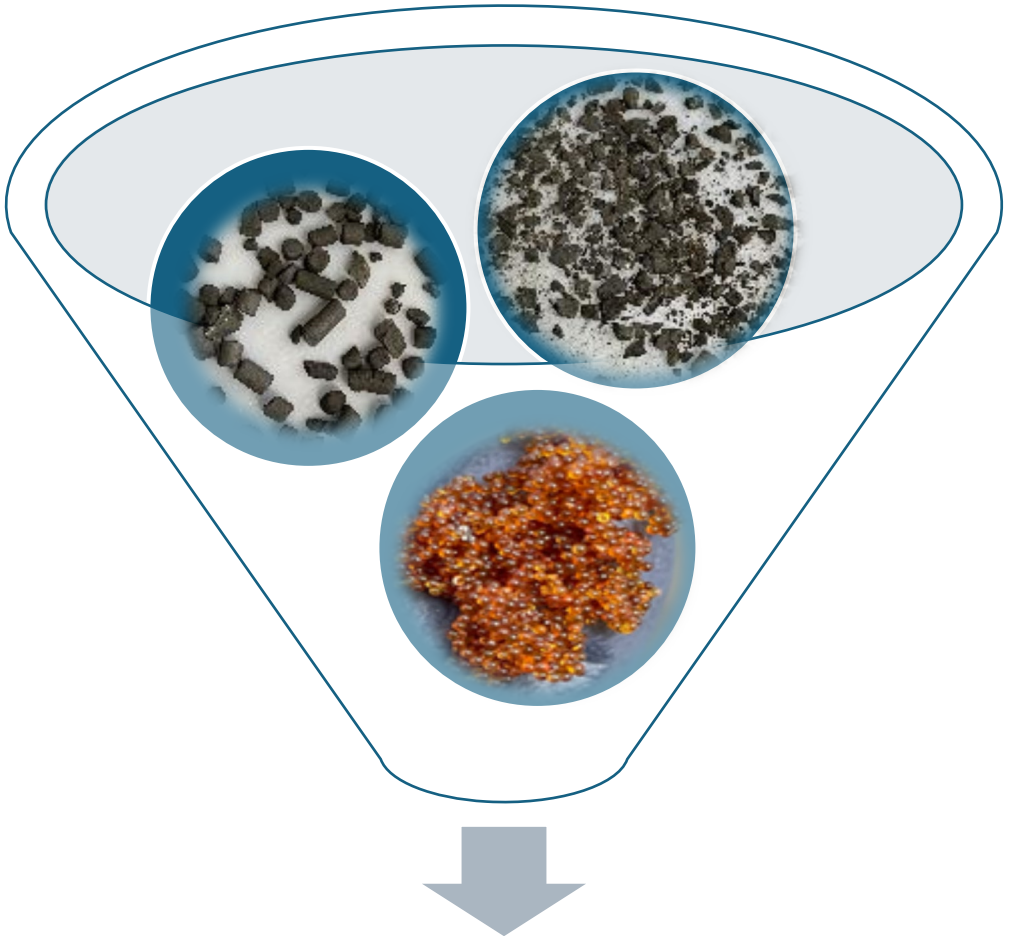
© 2025 AxNano Inc

A FEED MANAGEMENT SYSTEM IS NEEDED TO INCREASE UTILITY OF SCWO FOR THE INDUSTRY

Project Goal: Demonstrate that AxNano's SCWO with feed management system will create value for Air Force hazardous substances treatment beyond SCWO treatment alone.

What it is:
- Solid + Liquid waste processing system
- Automated controls

What it does:
- Enables SCWO continuous operation with varying wastes
- Reduces particle sizes to a desired particle size by wet or dry grind
- Meters and mixes different waste streams and additives
- Homogenizes feed

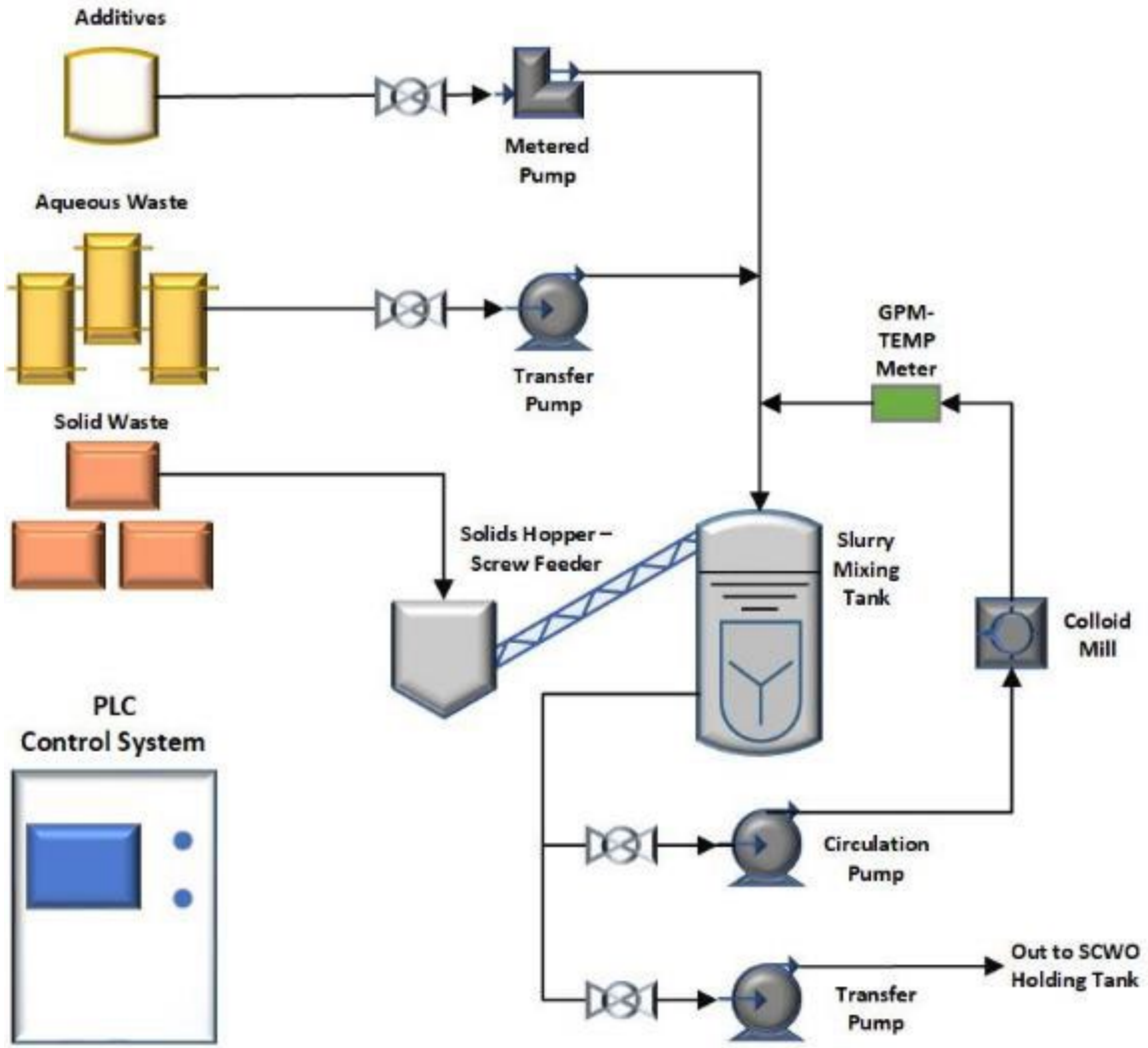


iSCWO

© 2025 AxNano Inc

Contract: FA864923P1125

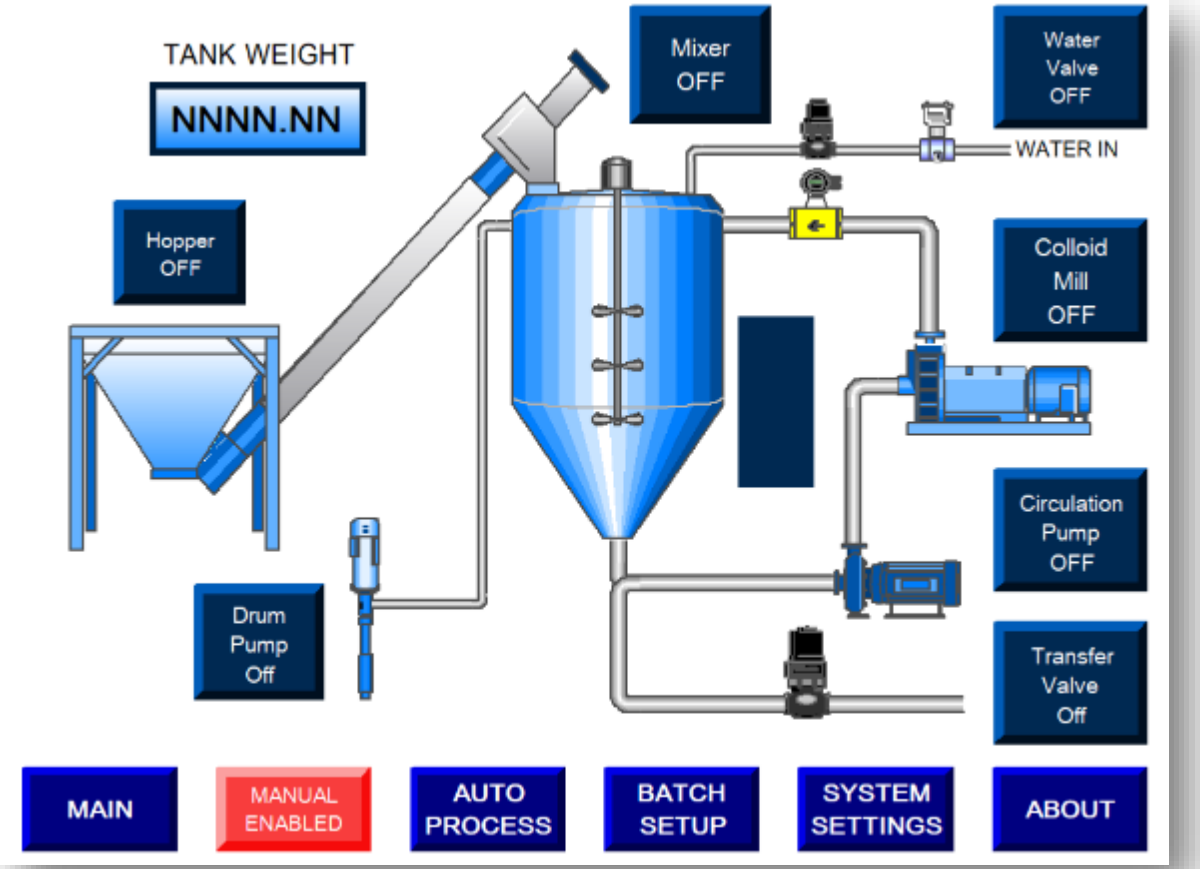
AUTOMATIC FEED SYSTEM (AFS) OVERVIEW



Project Objective: Design and test AFS prototype equipment that can produce a feed that meets known boundary conditions for General Atomics's iSCWO. Seven studies were conducted.

© 2025 AxNano Inc

AFS FEATURES FEED PROCESSING AND CONTROLS, WHILE DECREASING HUMAN EXPOSURE TO WASTES



Human Machine Interface for PLC

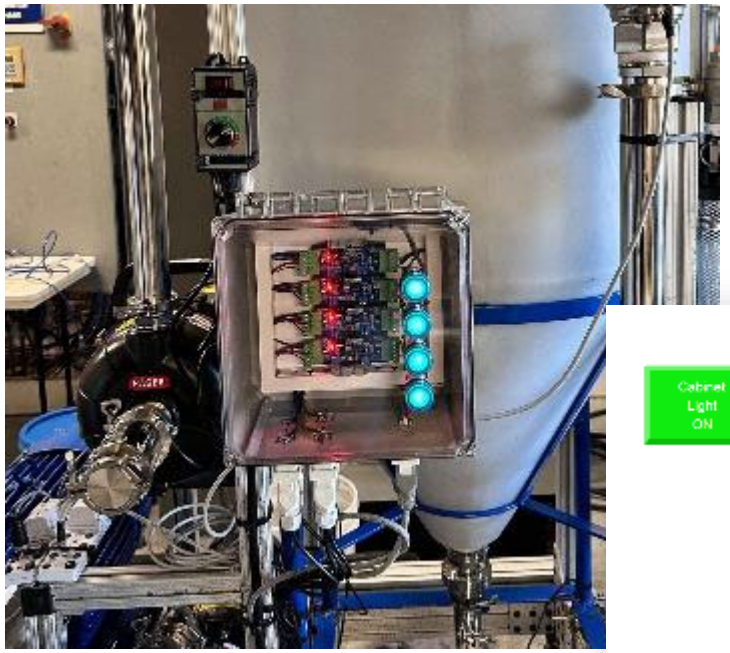
- PLC successfully controlled machine
- Replaces manual operations

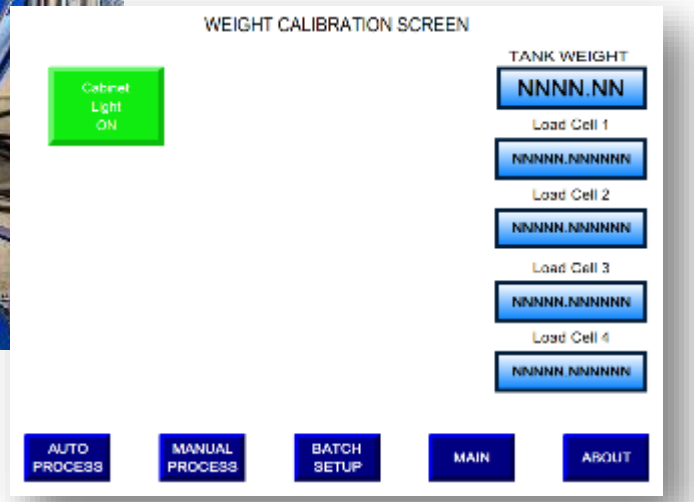
Benefits:

- Increases safety
- Protects components
- Increases throughput to SCWO

© 2025 AxNano Inc

AFS FEATURES FEED PROCESSING AND CONTROLS, WHILE DECREASING HUMAN EXPOSURE TO HANDLING WASTES





Human Machine Interface for Load Cells

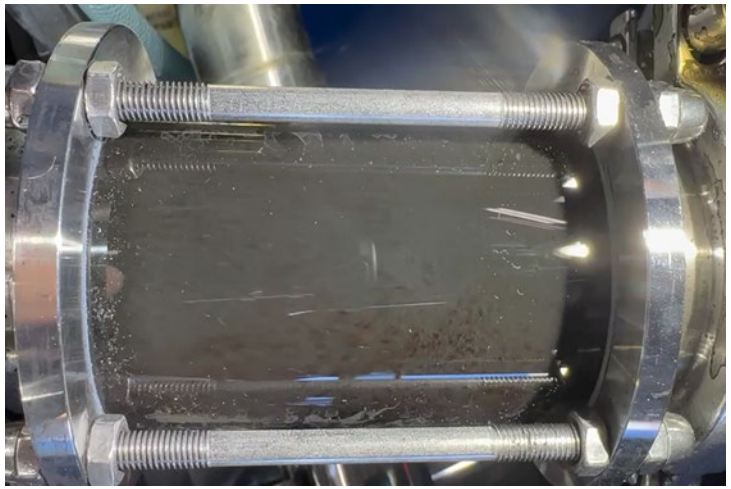
- PLC successfully controlled machine
- Replaces manual operations

Benefits:

- Increases safety
- Protects high cost components
- Increases throughput

© 2025 AxNano Inc

AFS FEATURES FEED PROCESSING AND CONTROLS, WHILE DECREASING HUMAN EXPOSURE TO HANDLING WASTES



- Meters flow rate
- Measures temperature

Benefits:

- Increases safety
- Protects high cost components
- Improves controls

© 2025 AxNano Inc

AFS PROCESSED IX AND GAC TO MEET iSCWO BOUNDARY CONDITIONS

GAC2 Particle Size:

< 2000 µm

AFS

GAC2 Particle size:

< 250 µm

90% of GAC particles were ground from 2mm to < 45 microns, while the remaining 10% were <186 microns, well below the target particle size boundary condition of 250 microns.

Resin Particle Size:

> 600 µm

Resin Particle size:

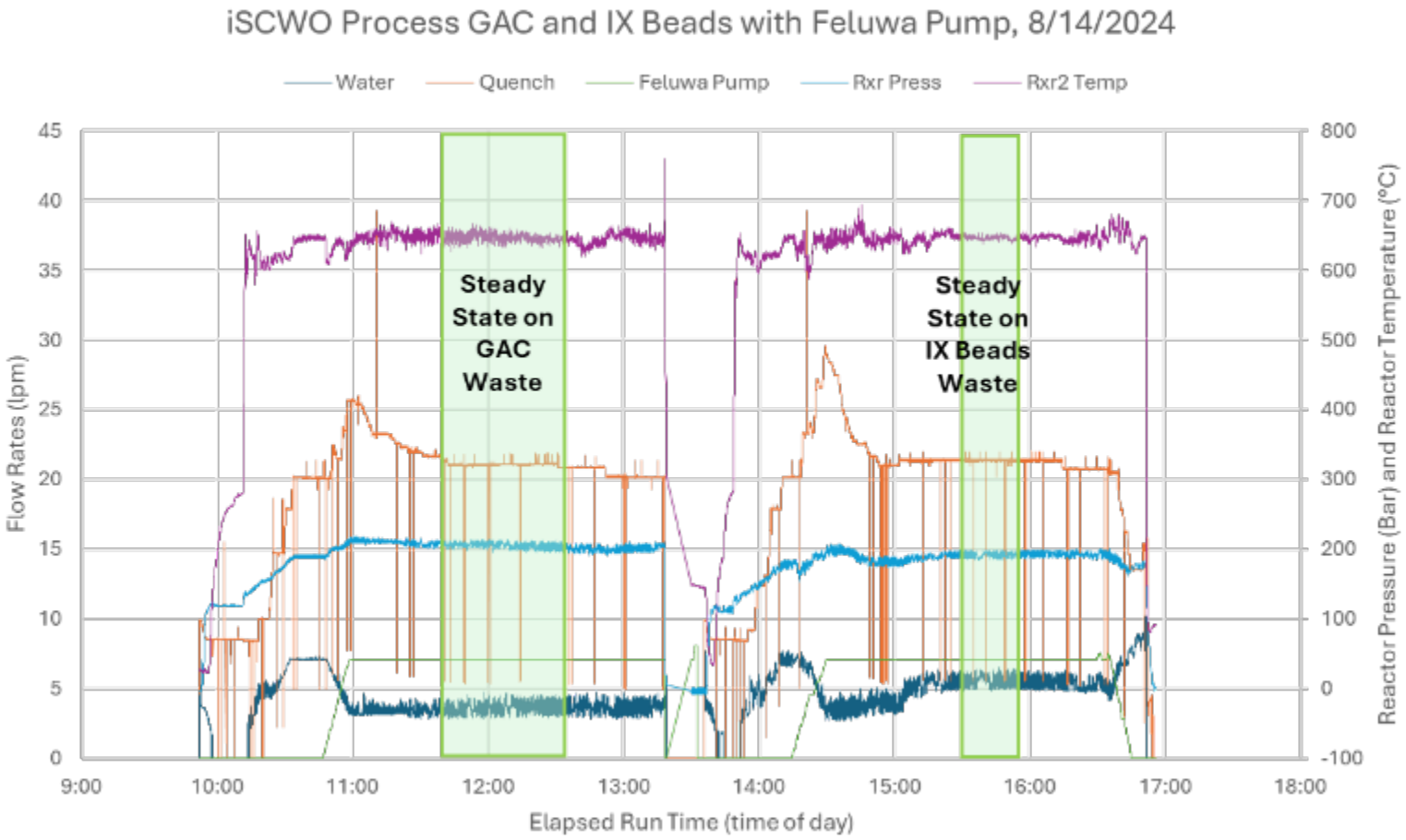
< 500 µm

60% particles of the total volume were <250 microns, with the remaining 40% ranging between 250 and 1,110 microns.

© 2025 AxNano Inc

ADDITIONAL TESTING ON IX FURTHER REDUCED PARTICLE SIZES

AFS PROCESSED IX AND GAC SUPPORTED CONTINUOUS iSCWO OPERATION

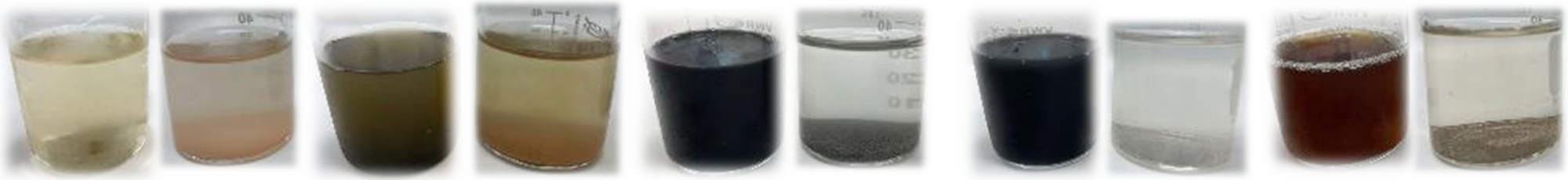


[1] Volume processed was determined by programmatic constraints, not equipment constraints. Larger volume testing will continue on ESTCP/DIU project HQ0034-23-9-0014 with General Atomics, Bay West and AxNano.

© 2025 AxNano Inc

UP TO 98% REDUCTION IN SOLIDS FOR SLURRIED WASTE STREAMS

Waste	Percent Solids Influent	Percent Solids effluent	Solids Reduction	Observations
Biosolids	10%	0.6%	94%	Brownish particles
AFFF wastewater	0.01%	1.0%	NA	White salt-like particles
RO reject	2%	0.5%	15%	Yellow green particles
GAC	10%	0.4%	96%	White salt-like particles
Resin	10%	0.2%	98%	Brownish particles



AFFF Biosolids GAC Resin RO Reject

© 2025 AxNano Inc

AFS-PROCESSED SOLIDS ACHIEVED PFAS REMOVAL SIMILAR TO LIQUIDS

Waste	Influent Total PFAS (ng/L)	[a]Normalized Effluent Total PFAS (ng/L)	Destruction Removal Efficiency
Baseline	-	242 (effluent concentration)	-
GAC	1,115,224	50.4	99.995%
Resin	134,112[b]	1,991.3	98.52%[b]
Biosolids	4,969.7	62.1	98.57%
AFFF wastewater	10,124	414.4	95.91%
RO Reject	13,443	45.4	99.66%

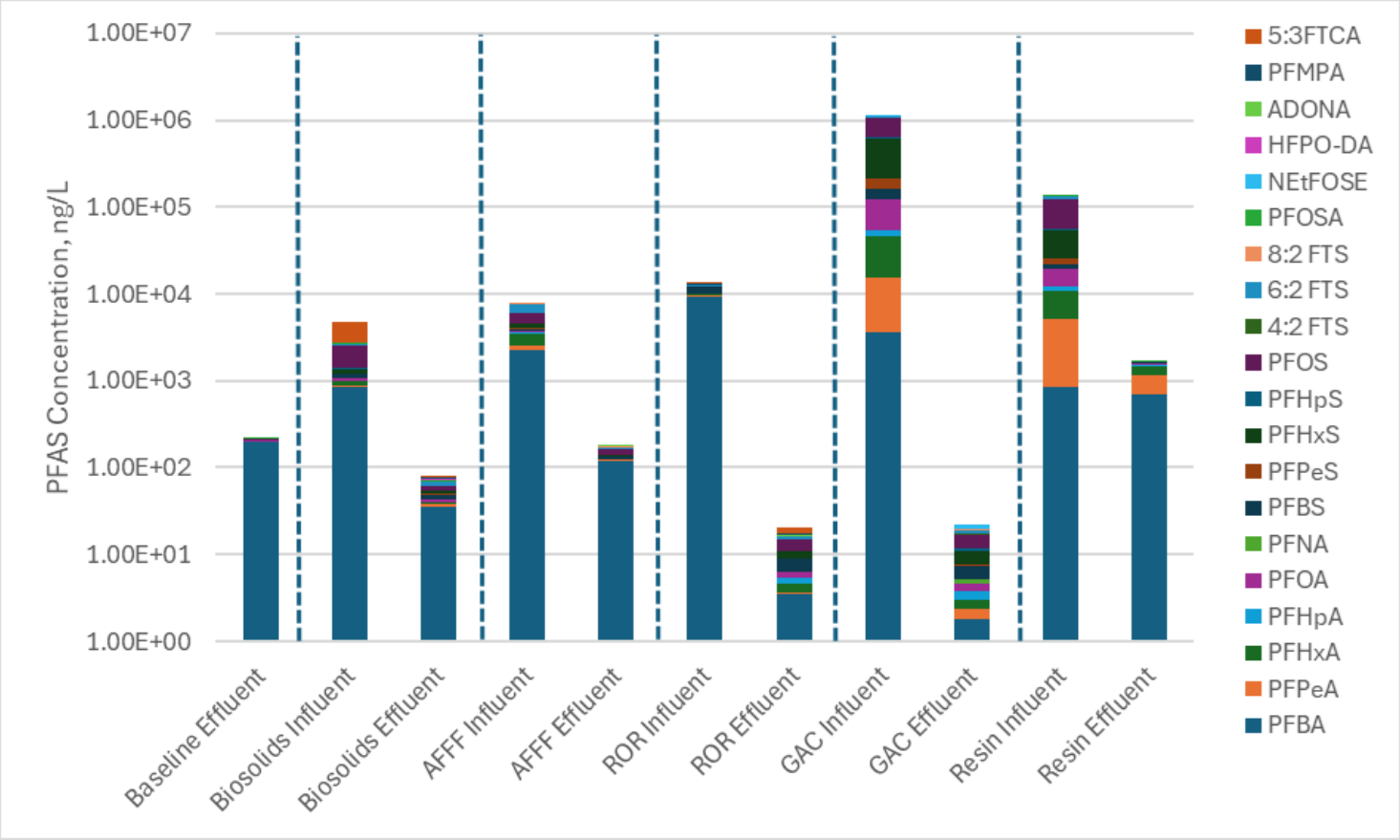
[a] Normalized to account for effluent dilution from process and quench water added. [b]Starting concentration of PFAS on resins may be inaccurately low

As measured by EPA Method 1633

- n=1 for each, unoptimized parameters.
- Longer run testing on ESTCP/DIU project HQ0034-23-9-0014 with General Atomics, Bay West and AxNano.

© 2025 AxNano Inc

AFS-PROCESSED SOLIDS ACHIEVED PFAS REMOVAL SIMILAR TO LIQUIDS



As measured by EPA Method 1633

- Presence of solids did not impact PFAS destruction
- Influent sampled before slurry pump inlet, effluent sampled after neutralization tank
- Background PFAS suspected from quench water (post-reactor)
- Need improved resin starting concentration analysis
- Optimized parameters (Bay West) will be used for full scale destruction on DIU/ESTCP project Q2 2025

© 2025 AxNano Inc

TOTAL ORGANIC CARBON AND CHEMICAL OXYGEN DEMAND WERE SIGNIFICANTLY REDUCED IN BOTH LIQUID AND SOLID WASTES

Waste	COD (mg/L)		COD Removal Efficiency	TOC (mg/L)		TOC Removal Efficiency
	Inf	**Eff**	**%**	**Inf**	**Eff**	**%**
Baseline	NA	NA	NA	NA	5.5	NA
Biosolids	11,060	34	99.69	36	9	73.89
AFFF wastewater	720	38	94.69	46	34	26.09
RO reject	143,100	35	99.98	46,000	8	99.98
GAC	14,980	1,077	92.81	710	4	99.46
Resin	10,290	58	99.44	290	3	99.44

AxNano

© 2025 AxNano Inc

OVERVIEW OF WASTE AND PROCESS PARAMETERS – SOLIDS AND LIQUIDS

	GAC	Resin	Biosolids	AFFF wastewater	RO reject
Volume Processed[1]	150 gal	90 gal	500 gal	55 gal	650 gal
PFAS Removal Efficiency (%)	99.995	98.52	98.75	95.91	99.66
Energy Content (BTU)	20,000	880	1,800	ND < 1	4,400
COD (mg/L)	14,980	10,290	1,1060	720	143,100
TOC (mg/L)	710	290	36	46	46,000
Feed Rate (lpm)	7.0	7.0	7.0	1.0	8.2
Reactor Pressure (bars)	204.8	191.9	215.8	227.9	200.1
Salt (mg/L)	257	268	99	2.7	2,508
Influent PFAS Concentration (µg/L)	1,115.2	134.1	5	10.1	13.4
Reactor Temperature (C°)	646±5	647±2	645±5	649±12	646±4
90% Particle size (µm)	< 51.8	< 859	< 756	NA	NA
Solids wt%	10%	15%	10%	NA	NA
Approx. Waste: diesel: water ratio	7:0:4	7:0:6	4:1:2	1:1:4	14:1:1

[1] Volume processed was determined by programmatic constraints, not equipment constraints. Larger volume testing will continue on ESTCP/DIU project HQ0034-23-9-0014 with General Atomics, Bay West and AxNano.

© 2025 AxNano Inc

COST ANALYSIS – SCOPE AND METHODS

Scope: Preliminary cost model of the AxSure™ system (AFS+iSCWO) on-site compared to off-site incineration and landfilling

Objective:

- Provide comparisons of AxSure™ as government-owned, contractor-operated compared to off-site landfilling and incineration
- Three scenarios to represent the various wastes processed in pilot test
- Test the value proposition of Automated Feed Management System (AFS)

Data inputs:

- Actual parameters from testing solid and liquid waste streams
- AFCEC, Base Environmental Flights, and Contractors for current landfilling costs.
- EPA interim guidance document for incineration because no current DoD comparisons exist

© 2025 AxNano Inc

COST ANALYSIS – ASSUMPTIONS

Assumptions:	
Gallons Waste Processed by AxSure™ Per Year[a]	1,296,000
Tons Waste Treated per Year - Scenario 1 (10% solid waste and 90% clean water)	172
Tons Waste Treated per Year - Scenario 2 (10% solid waste and 90% wastewater)	3512
Tons Waste Treated per Year - Scenario 3 (100% wastewater)	3900
Landfill Cost Per Ton[c]	$ 840
Incinerator Cost Per Ton[d]	$ 2,822
Waste Transportation Costs Per Ton[e]	$ 706

a 2.90 gallons per minute waste throughput based on August testing of RO reject. Year round operation with 27% down time (running 300 out of 360 day operation based on estimate from consolidated waste processing facility operator estimates and currently operating GA-made SCWO systems)

b

c From actual disposal cost from a federal facility, landfill disposal of resin, not including transportation

d From 2025 INTERIM GUIDANCE ON THE DESTRUCTION AND DISPOSAL OF Perfluoroalkyl and Polyfluoroalkyl Substances and Materials Containing Perfluoroalkyl and Polyfluoroalkyl Substances— Version 2. Does not include transportation. (Appendix D page D-2)

e From actual disposal costs from federal facility ANG GAC and resin roll-offs disposal

© 2025 AxNano Inc

COST ANALYSIS –OPERATING COSTS CAN BE LOWERED BY MANAGING FEED

- Inputs: BTU content, flow rate, utilities, other consumables, labor
- Mixing of solids and liquids drives down costs by increasing throughput
- Operating costs estimated $0.33/gal - $0.70/gal

	Description	Normalized Cost per ton to Scenario 1
Scenario 1	10% GAC slurry from clean water	1
Scenario 2	10% GAC slurry from wastewater	-93%
Scenario 3	100% Wastewater	-86%

© 2025 AxNano Inc

ON-SITE SCWO CAN OFFER COST SAVINGS COMPARED TO OFF-SITE DISPOSAL

		Estimated Cost Benefit (%) of onsite SCWO vs Off-Site Other over 10 Years
Purchase AxSure™ Equipment		
Scenario 1 (10% solids and clean water)	Total Annual Landfill Cost	-213%
	Total Annual Incinerator Cost	-37%
Purchase AxSure™ Equipment		
Scenario 2 (10% solid and 90% wastewater)	Total Annual Landfill Cost	83%
	Total Annual Incinerator Cost	93%
Purchase AxSure™ Equipment		
Scenario 3 (100% wastewater)	Total Landfill Cost	78%
	Total Annual Incinerator Cost	90%

- For customers generating enough wastewater to support continuous operation, a government-owned/contractor-operated on-site SCWO can save up to 83% compared to off-site landfilling or 93% compared to incineration
- For customers only generating solids PFAS wastes, landfilling is the most economic method, though longterm liability is not considered in this model.

© 2025 AxNano Inc

PROJECT CONCLUSIONS

- Demonstrated ability to process solids not previously processable on iSCWO
- AFS-prepared slurries from spent media supported stable iSCWO operation
- AFS throughput meets the requirements for the continuous operation of the SCWO system
- Cost reductions may be available for on-site versus off-site for customers generating mixed solid+liquid waste streams

© 2025 AxNano Inc

Come visit! See the system in action at Charlotte NC

Stay tuned! Look for our papers on additional results and case studies

Got solids? Contact us
alexis.carpenter@axnano.com

AxNano

© 2025 AxNano Inc